

Course Guide for Physics 131: Electromagnetic Theory I

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(Dated: November 19, 2025)

I. COURSE INFORMATION

Description: Electrostatics in a vacuum, electrostatics in dielectric media, boundary value methods in electrostatics, electric currents, conducting media, magnetostatics in a vacuum, macroscopic and microscopic magnetism, Faraday's law of electromagnetic induction.

References: [Griffiths](#) [GR]

Prerequisites: Physics 107 and Physics 117

Credits: 3.0

Schedule: 2× weekly lecture meetings, 1.5 hours each.

Course webpage: sand.nip.upd.edu.ph/physics131

Class communication: Group email (I have added the email address you supplied in your CRS profile). Please let me know if you are not receiving messages through the group.

II. OVERVIEW

This course pack consists of guided readings and exercises throughout the semester. It is important that students acquire the competencies listed in this guide.

Any reference text or online reference on undergraduate-level electromagnetism is sufficient to cover the material in this course. The outline content follows [Griffiths](#) [GR]. Some of complementary references are available for download from the University Library.

III. ONLINE RESOURCES

The University Library has access to downloadable books on electromagnetic theory. Please use your @up.edu.ph credentials to access books from the [Library eBooks databases](#) via an OpenAthens log in.

[MIT OpenCourseWare](#) offers free online course materials for classes in physics under a particularly relaxed Creative Commons license. An online resource that may

help is OCW Resource textbook [Electromagnetic Fields and Energy](#). A class with a similar coverage to ours is the MIT course [Physics 8.07 Electromagnetism II](#).

IV. STUDY AND ACTIVITY GUIDE

The syllabus serves as a study and activity guide: the weekly learning outcomes can be achieved by (1) studying and reading the corresponding learning resource and consulting other equivalent resources, (2) doing the in-text examples and sample exercises in the course pack, and (3) performing the items in the assignment guide. Although the items are listed on a weekly basis, the activities can be done at your own pace as long as they are submitted by the deadline ■. In these times we are expected to learn independently, but do not hesitate to ask me for help on anything. You can reach me by email, phone, and on our online platform if you need any assistance.

V. CLASS REQUIREMENTS AND ASSIGNMENT GUIDE

You must submit two (2) notebook files, one (1) problem set, and take one sit-in midterm exam to complete the course. Notebooks and problem sets may be handwritten or typeset, but the submission must be a PDF file to be submitted through an online form. The submission link will be provided on the course webpage. You must submit your work by the deadline, otherwise the work will be considered a non-submission. The submission bin will close 48 hours after the deadline as a grace period, and late submissions within this period must be accompanied by a University-recognized excuse.

On-site attendance will be checked to comply with University attendance requirements.

Progress will be monitored by performing two types of checkpoint activities:

1. □ Notebook entries

You must submit two notebook files (one before midterm, and another before the finals week). Each file must contain weekly notebook entries with a new section or page made for each week. Each entry should demonstrate that each box-bulleted □ outcome item in the syllabus has been achieved. For example, you can write a short summary or

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outline of the topic, or give a list of steps in the derivation of a core result, or provide a worked out representative example. I suggest that you monitor your progress by ticking each box ☐ in this course guide as you complete these notebook entries.

By the end of the semester you will have a notebook that you can use as a handy reference as you progress in the BS Physics/BS Applied Physics program.

2. ☆ Problems

You must submit one problem set to be given later in the semester. You are encouraged to discuss these problems with your classmates and help each other, but you **must work independently** and write up your own reports and solutions. You can also ask me for help. It is recommended to do the problems in parallel with the learning units, but you can work on them in any order and at your own pace.

VI. CLASS POLICIES

Announcements: Class announcements will be made on the [class webpage](#) and/or by email.

Availability: You may email me for inquiries, but please make an effort to seek answers yourself before doing so.

Assessment: There will be one problem set, one midterm exam, and two notebook submissions required in this class. Each problem set, exam, and notebook submission will contribute equally to the final grade. The problem set, midterm exam, and notebook files must be submitted: a grade of INC

will be given if one of the four requirements are not submitted in class (for the midterm) or via the submission bin; a grade of 5.0 will be given if two or more requirements are not submitted. At most one non-submission or 0-graded requirement may be replaced by taking a comprehensive written final examination during the Finals week.

On-site attendance will be checked to comply with University attendance requirements.

Academic honesty: You are encouraged to collaborate with your classmates and use reference texts to help you with your problem sets. However, you must submit solutions that you have written independently.

College of Science statement: Any form of cheating in examinations or any act of dishonesty in relation to studies, such as plagiarism, shall be subject to disciplinary action.

Raw score x	Point grade
$90\% \leq x \leq 100\%$	1.00
$85\% \leq x < 90\%$	1.25
$80\% \leq x < 85\%$	1.50
$75\% \leq x < 80\%$	1.75
$70\% \leq x < 75\%$	2.00
$65\% \leq x < 70\%$	2.25
$60\% \leq x < 65\%$	2.50
$55\% \leq x < 60\%$	2.75
$50\% \leq x < 55\%$	3.00
$45\% \leq x < 50\%$	4.00
$x < 45\%$	5.00

VII. SYLLABUS WITH EXPECTED WEEKLY OUTCOMES

First day of classes : 21 Aug 2024

W01 13-15 Aug : Orientation and introduction

W02 20-22 Aug : Review of vector calculus

Reference: GR 1

- ☐ Provide geometrical interpretations or physical examples for the gradient, divergence, and curl. Make a sketch or add a plot that you have made to illustrate these (point) derivatives.
- ☐ Provide geometrical interpretations or physical examples for the fundamental theorem for gradients, divergences, and curls.
- ☐ Write down Laplacian for an orthogonal coordinate system in terms of its scale factors. Verify this formula for cylindrical and spherical coordinates.

- Prove that the divergence of a radial inverse-square power vector field is consistent with a Dirac-delta point source or sink.

W03 27–29 Aug: Electrostatics: The electric field
References GR 2

- Write down the expression for the electric field $\mathbf{E}(\mathbf{r})$ due to a continuous charge distribution $\rho(\mathbf{r})$. Use sketches that show how the field $\mathbf{E}$ at the field point $\mathbf{r}$ is integrated over differential elements about source points $\mathbf{r}'$.
- Write down and give a brief physical description for the Maxwell's equations that involve the divergence and curl of $\mathbf{E}$.
! For charge distributions $\rho(\mathbf{r})$ that are symmetric, the construction of Gaussian surfaces with the same symmetry allows one to readily obtain electric fields by the divergence theorem.
! When charge distributions are constant in time or when current distributions are constant in time, the curl of the associated electric field is zero.
- Consider a dipole with charge q with separation distance a and dipole moment $p = qa$. Evaluate the electric potential and electric field due to this configuration. Then take the point dipole limit $a \rightarrow 0$ with p finite. Find the far-field electric potential and electric field.

W04 3–5 Sep: Electrostatics: Boundary conditions, work and energy, conductors
References GR 2

- Give the continuity conditions on the electric field and potential across interfaces.
- Calculate the work needed to assemble a uniformly charged sphere of charge density ρ_0 and radius R from charges at infinity.
- Provide the features of conductors that are relevant to determining the distribution of charges in them at electrostatic equilibrium.
- Briefly discuss how induced charges are set up on the surfaces of conductors in the presence of external electric fields.

W05 10–12 Sep: Calculating scalar potentials I
References GR 3

- Summarize the conditions needed to guarantee a unique solution for Laplace's equation.
- Write down the general solution to Laplace's equation as an eigenfunction expansion in Cartesian and spherical coordinates.

W06 17–19 Sep: Calculating scalar potentials II
References GR 3

- Discuss the main steps on how to write a scalar potential as a multipole expansion.
- Give the first three terms of the multipole expansion (up to quadrupole order) and give a brief physical description of each. Write the general expression for a multipole as a general moment of the charge density.
- Use a multipole expansion to obtain the leading non-vanishing term for the far-field ($a/r \ll 1$) potential for a dipole of charge q , separation distance a .

W07 24–26 Sep: Dielectrics
References GR 4

- ! Dipole moments in matter may be induced or permanent.
- ! A polarization field $\mathbf{P}$ gives the total density of dipole moments in matter (induced and permanent).
- ! This polarization field leads to effective “bound” surface $\sigma_b = \mathbf{P} \cdot \hat{\mathbf{n}}$ and volume charge densities $\rho_b = -\nabla \cdot \mathbf{P}$.

Midterm Exam : Friday, 03 Oct 2025

W08 08–10 Oct: Electric displacement

References GR 4

- Make a clear distinction between the conventional “free” and “bound” charges described in the textbook.
- Show how Gauss’s law is modified in the presence of a polarization field $\mathbf{P}$. Make sure to label with complete physical descriptions each term in the definition of the electric displacement field $\mathbf{D}$.
- Describe a linear dielectric in terms of the relationship between the polarization and electric fields.
- Give the boundary conditions (normal and tangential) on the electric field $\vec{E}$ and displacement field $\vec{D}$ at a boundary between two dielectric media.

W09 15–17 Oct: Energy in dielectrics, Magnetostatics

References GR 4

- Show that the work needed to charge a dielectric capacitor, and hence the energy stored in it, increases with the relative permittivity of the dielectric.

References GR 5

- Explain why a single moving point charge does not constitute a steady current.
- GR Ex. 5.6: Use the Biot-Savart law to obtain the magnetic field at a distance z above the center of a circular loop of radius a that carries a steady current I . Give the result for the two possible directions of the current flow.

W10 22–24 Oct: Magnetic vector potential

References GR 5

- Demonstrate that two vector potentials that differ by the gradient of the function yield the same magnetic field. What is this invariance called?
- Calculate the vector potential of an infinite sheet of uniform and constant current density (direction and magnitude). Verify that the magnetic field is consistent with the Biot-Savart/Ampère’s law.

Deadline ■ : Notebook 1 (Weeks 1–9) due 24 Oct 2025.**Deadline ☆ : Problem Set 1 due 24 Oct 2025.**

W11 29 Oct: Magnetostatics

References GR 5

- Make a table showing the analogous electrostatic and magnetostatic relationships between fields and source densities, Maxwell’s source equations in integral and differential forms, and Laplace’s and Poisson’s equations for the field potentials.
- ! The tangential component of the magnetic field is discontinuous across an interface with a surface current, while its component normal to the surface is continuous.
- Obtain the far-field vector potential and magnetic field due to a circular circuit of radius a carrying a steady current I . Show that the effective magnetic dipole moment of this circuit at large distances is equal to the product of the current and the area enclosed by the circuit.

W12 05–07 Nov: Magnetic fields in matter

References GR 6

- Make a table comparing induced electric and magnetic polarization in matter.
- Differentiate between bound and free current sources in matter, and between paramagnets, diamagnets, and ferromagnets.
- Differentiate between the $\vec{B}$ and $\vec{H}$ fields.
- Give the boundary conditions that the $\vec{H}$ field must satisfy at the interface between two materials.

W13 12–14 Nov: Boundary value problems in magnetostatics

References GR 6

- GR 6.16: Consider a coaxial cable consisting of two long concentric cylindrical conductors of radii $0 < a < b$ and negligible thickness. The material between them is an insulator with constant magnetic susceptibility χ_m . A current I flows down the inner tube and up the other. The current density on the conductors is uniform for each tube. Find the magnetic field $\mathbf{B}$ in between the two tubes. Verify your answer by computing the magnetization and bound current density and check that the fields add up together correctly.
- GR 6.16 additional: In the previous situation, there is no magnetic material inside the inner tube and outside the outer tube. Obtain the magnetic field $\mathbf{B}$ inside the inner cylinder and outer cylinder (can you use Ampère's circuital law here?). Show that the boundary conditions on the discontinuities of the $\mathbf{H}$ field are consistent with your solution.

W14 19–21 Nov:

References GR 6

- Calculate the magnetic vector potential due to a uniformly magnetized sphere.
- Show that the magnetic field outside the sphere has the functional form of the field of a point dipole.

W15 26–28 Nov:

References GR 6

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Deadline ■ : Notebook 2 due 05 Dec 2025.

Integration period : 06 Dec 2025

Finals week : 09–16 Dec 2025

REFERENCES

D. J. Griffiths, *Introduction to Electrodynamics* (Pearson Education, Inc., 2013), 4th ed.